**P 5.2 Analytical Procedures**

Enclosed Herewith

| | | | |
|---|---|---|---|
| LEBEN LIFE SCIENCES | Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA | STP No. | LT010/FP/STP/042/01 |
| | | Supersedes | 00 |
| Quality Control Department | Finished Product Standard Test Procedure | Effective Date | 25/01/2023 |
| | Product Name | Solifenacin Succinate Tablets BP 5 mg | |
| | Reference | BP/IH | Market | Export |

1. **Description:**
   **Instruments** :Analytical Balance
   **Reagents** :NA
   **Procedure** : Take about 10 tablets into a clean and dry Petri-dish and observe.

2. **Identification:**
   **A. By IR:**
   **Instruments** : FT-IR
   **Reagents** : Sodium Hydroxide, dichloromethane, sodium sulphate.
   **Procedure:**Disperse an amount of powdered tablets containing 20 mg of Solifenacin Succinate in 30 mL of water and filter (a 0.45-µm Nylon filter is suitable). To this solution add 5 mL of 5M sodium hydroxide. Extract with three 30-mL quantities of dichloromethane, shake the combined extracts with anhydrous sodium sulfate, filter and evaporate the filtrate to dryness. The infrared absorption spectrum of the residue is concordant with the reference spectrum of solifenacin.

3. **Average Weight:**
   **Instruments** : Analytical Balance
   **Reagents** : NA
   **Procedure** : Take a weight of 20 tablets individually and calculate the average weight by using following formula.

$$\text{Average Weight} = \frac{\text{weight of 20 tablets}}{20}$$

4. **Uniformity of Weight:**
   **Instruments:** Analytical Balance.
   **Reagents** : NA
   **Procedure:**
   1. Take a weight of 20 intact Tablets individually (W1)
   2. Determine the average weight of Tablets using following formula.

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$$\text{Average weight} = \frac{\text{sum of weight of individual Tablet}}{20}$$

3. Calculate the 7.5 % of average weight using following formula(z)

$$Z = \frac{\text{Average weight}}{100} \times 7.5$$

4. Calculate the limit for uniformity of weight using following formula
   Higher limit = average weight + Z
   Lower limit = average weight - Z

5. Check the weight of individual Tablet within above limit.

**5. Content Uniformity:**

**Instruments** :Analytical Balance, HPLC, pH meter.

**Reagents** :PotassiumDihydrogen orthophosphate monohydrate, Acetonitrile, Orthophosphoric Acid, Methanol.

**Procedure:**

Select 10 tablets randomly. Weigh tablet individually. Calculate by preparing and injecting 10 sample solutions.

**Preparation of Buffer Solution:** Dissolve 3.4 g of potassium dihydrogen orthophosphate in 1000 mL water and adjust the pH to 3.5 with 85% v/v orthophosphoric acid.

**Preparation of Mobile Phase:**400 ml of buffer solution and 600 ml of a solution containing 200 ml of water, 400 ml of acetonitrile and 400 ml of methanol.

**Sample solution (Solution 1):**

Disperse 1 tablet in each 50 ml volumetric flask, add 30 ml of mobile phase and sonicate for 20 minutes to dissolve completely, dilute upto mark to produce 50 ml with same mobile phase, filter with 0.45µm nylon filter by discarding first 1 ml.

**Standard Solution(Solution 2):** Weigh and transfer 50 mg of Solifenacin Succinate WS in 50 ml of volumetric flask, add 25 ml of mobile phase and sonicate to dissolve completely,

make up the volume up to mark with same mobile phase. Dilute 5 ml of this solution to 50 ml with mobile phase.

**Chromatographic conditions:**

| | |
|---|---|
| **Column** | Stainless steel column (25 cm × 4.6 mm) packed with cyanopropyl silica gel for chromatography (5 µm) (Agilent Zorbax SB-CN is suitable) or equivalent. |
| **Injection volume** | 20µl |
| **Column Temperature** | 35°C |
| **Detection** | 220 nm |
| **Flow rate** | 1.0 ml per min |
| **Run time** | 15 Minutes |

**Sample Sequence:**

| Solution | No. of injections |
|---|---|
| Blank | 1 |
| Standard solution | 5 |
| Sample solution | 1 of Each |
| Bracketing Standard solution | 1 |

**System Suitability Criteria: (Standard Solution)**

% RSD for area of the standard five replicate injections is NMT 2.0%.

**Determination of Content:**

Calculate the content of solifenacin succinate, $C_{27}H_{32}N_2O_6$, in the tablets from the chromatograms obtained and using the declared content of $C_{27}H_{32}N_2O_6$ in solifenacin succinate RS/WS.

$$\%\text{Assay} = \frac{Au}{As} \times \frac{Wt.Std}{50} \times \frac{5}{50} \times \frac{100}{Wt.Spl} \times \frac{Avg.wt}{LC} \times \text{Potency}$$

Where,

Au :- Peak area obtained due to sample solution

As :- Peak area obtained due to standard solution

Wt. Std :- Weight of Standard

Wt. Spl :- Weight of Sample
Avg. Wt :- Average Weight
LC :- Label Claim

Calculate AV value for each component using following formula.

$$AV = |M - X| + ks$$

Where,
X = Mean of individual content
M = Reference Value
k = Acceptability constant
If n=10 then k = 2.4
If n=30 then k = 2.0
s = Sample standard deviation

**6. Hardness:**

**Hardness:**

**Instruments** : Hardness tester

**Reagents** : NA

**Procedure:** Place the tablet between the jaws of hardness tester, taking into account, where applicable, the shape, the break-mark and the inscription; for each measurement orient the tablet in the same way with respect to the direction of application of the force. Carry out the measurement on 10 tablets, taking care that all fragments of tablets have been removed before each determination.

**7. Friability:**

**Instruments** : Analytical Balance, Friability apparatus

**Reagents** : NA

**Procedure:** The tablets are carefully de-dusted prior to testing. Accurately weigh 6.5 g of the tablet sample, and place the tablets in the drum. Rotate the drum 100 times, and remove the tablets. Remove any loose dust from the tablets as before, and accurately weigh.

**Criteria:**

If the weight loss is greater than the targeted value, the test is repeated twice and the mean of the 3 tests determined. A maximum loss of mass (obtained from a single test or from the mean of 3 tests) not greater than 1.0 per cent is considered acceptable for most products.

$$\text{Friability} = \frac{\text{Initial Wt of tablets} - \text{Final Wt of tablets}}{\text{Initial Wt of tablets}} \times 10$$

8. **Disintegration time:**

**Instruments:** Disintegration test apparatus.
**Reagents** : NA
**Procedure:** Operate the apparatus for 30 minutes, note down the time required for complete disintegration of tablets.

9. **Dissolution:**

**Instruments:**Analytical Balance, Dissolution Test Apparatus, pH Meter, HPLC
**Reagents** :Hydrochloric Acid,Acetonitrile, PottasiumDihydrogen orthophosphate monohydrate, Orthophosphoric Acid, Methanol.

**Procedure**:
**Preparation of 0.1 M Hydrochloric Acid (Dissolution Medium):** Dissolve 8.5 ml of HCl in 1000 ml of water and mix well.

**Preparation of Buffer Solution:** Dissolve 3.4 g of potassium dihydrogen orthophosphate in 1000 mL water and adjust the pH to 3.5 with 85% v/v orthophosphoric acid.

**Preparation of Mobile Phase:**400 ml of buffer solution and 600 ml of a solution containing 200 ml of water, 400 ml of acetonitrile and 400 ml of methanol.

**Dissolution apparatus conditions:**

| | |
|---|---|
| Medium | 900 ml of 0.1M Hydrochloric Acid |
| Apparatus | Paddle |

| | |
|---|---|
| Temperature | 37.0±0.5°C |
| RPM | 50 rpm |
| Time | 30 Minutes |

**Preparation of Standard Stock Solution:** Dissolve 55 mg of Solifenacin Succinate WS in 100 ml of volumetric flask, add 70 ml of dissolution medium and sonicate to dissolve and dilute upto mark with the dissolution medium.

**Preparation of Standard Solution:** Further dilute 1 ml of standard stock solution to 100 ml with dissolution medium.

**Note:** Standard solution is stable up to 105 hours & Sample solution is stable up to 104 hours.

**Preparation of Sample solution:** Taken 1 tablet in each bowl containing 900 ml dissolution media, after 30 minutes withdraw 10 ml sample solution and filter with 0.45µm nylon filter by discarding 1 ml.

**Chromatographic conditions:**

| | |
|---|---|
| **Column** | Stainless steel column (25 cm × 4.6 mm) packed with cyanopropyl silica gel for chromatography (5 µm) (Agilent Zorbax SB-CN is suitable) or equivalent. |
| **Injection volume** | 20µl |
| **Column Temperature** | 35°C |
| **Detection** | 220 nm |
| **Flow rate** | 1.0 ml per min |
| **Run time** | 15 Minutes |

**Sample Sequence:**

| Solution | No. of injections |
|---|---|
| Blank | 1 |
| Standard solution | 5 |
| Sample solution | 1 of each |
| Bracketing Standard solution | 1 |

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When the chromatograms are recorded under the prescribed conditions, the retention time of solifenacin is about 6 minutes.

**System Suitability Criteria: (Standard Solution)**
%RSD for area of the standard five replicate injections is NMT 2.0%.

**Determination of Content:**
Calculate the total content of solifenacin succinate, $C_{27}H_{32}N_2O_6$ , in the medium from the chromatograms obtained and using the declared content of $C_{27}H_{32}N_2O_6$ in solifenacin succinate BPCRS.

$$\text{Dissolution (\%)} = \frac{AT}{AS} \times \frac{Ws}{100} \times \frac{1}{100} \times \frac{900}{1} \times \frac{1}{LC} \times P$$

Where,
AT = Peak area obtained in the sample solution.
AS = Peak area obtained in the standard solution.
Ws =Weight of the standard
LC = label claim.
P = potency

**10. Assay(By HPLC):**

**Instruments** : Analytical Balance, HPLC, pH meter.
**Reagents** : Potassium Dihydrogen orthophosphate monohydrate, Acetonitrile, Orthophosphoric Acid, Methanol.

**Procedure:**
**Preparation of Buffer Solution:** Dissolve 3.4 g of potassium dihydrogen orthophosphate in 1000 mL water and adjust the pH to 3.5 with 85% v/v orthophosphoric acid.
**Preparation of Mobile Phase:**400 ml of buffer solution and 600 ml of a solution containing 200 ml of water, 400 ml of acetonitrile and 400 ml of methanol.

**Note:** Standard solution is stable up to 80 hours & Sample solution is stable up to 23 hours.
**Sample solution (Solution 1):**
Weigh and powder 20 tablets. Transfer powder containing 10 mg of Solifenacin Succinate into 100 ml volumetric flask, add 50 ml of mobile phase and sonicate for 20 minutes to

dissolve completely, dilute upto mark to produce 100 ml with same mobile phase, filter with 0.45µm nylon filter by discarding first 1 ml.

**Standard Solution(Solution 2):** Weigh and transfer 50 mg of Solifenacin Succinate WS in 50 ml of volumetric flask, add 25 ml of mobile phase and sonicate to dissolve completely, make up the volume up to mark with same mobile phase. Dilute 5 ml of this solution to 50 ml with mobile phase.

**Chromatographic conditions:**

| | |
|---|---|
| **Column** | Stainless steel column (25 cm × 4.6 mm) packed with cyanopropyl silica gel for chromatography (5 µm) (Agilent Zorbax SB-CN is suitable) or equivalent. |
| **Injection volume** | 20µl |
| **Column Temperature** | 35°C |
| **Detection** | 220 nm |
| **Flow rate** | 1.0 ml per min |
| **Run time** | 15 Minutes |

**Sample Sequence:**

| Solution | No. of injections |
|---|---|
| Blank | 1 |
| Standard solution (Solution 2) | 5 |
| Sample solution_01 | 1 |
| Samplesolution_02 | 1 |
| Bracketing Standard solution (Solution 2) | 1 |

When the chromatograms are recorded under the prescribed conditions, the retention time of solifenacin is about 6 minutes.

**System Suitability Criteria: (Standard Solution)**

%RSD for area of the standard five replicate injections is NMT 2.0%.

**Determination of Content:**

Calculate the content of solifenacin succinate, $C_{27}H_{32}N_2O_6$, in the tablets from the chromatograms obtained and using the declared content of $C_{27}H_{32}N_2O_6$ in solifenacin succinate RS/WS.

$$\%Assay = \frac{Au}{As} \times \frac{Wt.Std}{50} \times \frac{5}{50} \times \frac{100}{Wt.Spl} \times \frac{Avg.wt}{LC} \times Potency$$

Where,

| | | |
|---|---|---|
| Au | :- | Peak area obtained due to sample solution |
| As | :- | Peak area obtained due to standard solution |
| Wt.Std | :- | Weight of Standard |
| Wt.Spl | :- | Weight of Sample |
| Avg.Wt | :- | Average Weight |
| LC | : - | Label Claim |

## 11. Related Substances By HPLC:

**Instruments** :Analytical Balance, HPLC, pH meter.

**Reagents** :Hydrogen Peroxide, Sodium MetabisulfateAcetonitrile, PotassiumDihydrogen orthophosphate, Methanol, Orthophosphoric Acid, Triethylamine.

**Procedure:**

**Preparation of Buffer Solution:** Dissolve 4.11 g potassium dihydrogen orthophosphate in 1000 mL water, add 1 mL triethylamine and adjust the pH to 3.3 with 85% v/v orthophosphoric acid.

**Preparation of Solution A:** 500 ml of buffer solution and 500 ml of acetonitrile.

**Preparation of Mobile Phase:**

**Mobile Phase A:** 150 ml of acetonitrile, 150 ml of methanol and 700 ml of buffer solution.

**Mobile Phase B:** 300 ml of buffer solution, 350 ml of acetonitrile and 350 ml of methanol.

**Gradient Program:**

| Time (Minutes) | Mobile Phase A | Mobile Phase B |
|---|---|---|
| 0-5 | 90 | 10 |

| | | |
|---|---|---|
| 5-25 | 30 | 70 |
| 25-40 | 10 | 90 |
| 40-55 | 10 | 90 |
| 55-60 | 90 | 10 |
| 60-65 | 90 | 10 |

**Preparation of 50 % w/v Sodium Metabisulfate:** Weigh and dissolve 5.0 gm of sodium metabisulfate in 10 ml of water.

**Note:** Standard solution is stable up to 57 hours & Sample solution is stable up to 51 hours.

**Preparation of Sample solution (Solution-1):**

Weigh and powder 20 tablets. Transfer powder containing 38 mg of Solifenacin Succinate into 25 ml volumetric flask, add 15 ml of solution A and sonicate to dissolve completely with occasional shaking and make up the volume up to the mark to produce 25 ml and filter with 0.45µm nylon filter by discarding first 1ml.

**Solution-2:**

Dilute 1 ml of sample solution (solution 1) to 200 ml.

**Solution-3:**To 15 mg of solifenacin succinate BPCRS/WS in 20 ml of volumetric flask, add 2.5 mL of solution A and 2.5 mL of hydrogen peroxide (30%), stir and then carefully add 2.5 mL of 50% w/v sodium metabisulfate and dilute to 20 mL with solution A (generation of impurity I).

**Solution-4:** Take 5 ml of standard solution (solution-2) in 25 ml volumetric flask and dilute to 25 ml with solution A.

**Solution-5:** Dissolve 1.52 mg of solifenacin succinate impurity BPCRS in 1 ml of solution A.

**Preparation of Placebo Solution:** Transfer placebo powder about 1135 mg into 25 ml volumetric flask, add 15 ml of solution A and sonicate to dissolve completely with occasional shaking and make up the volume up to the mark to produce 25 ml and filter with 0.45µm nylon filter by discarding first 1ml.

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| Quality Control Department | Finished Product Standard Test Procedure | | Effective Date | 25/01/2023 |
| | Product Name | Solifenacin Succinate Tablets BP 5 mg | | |
| | Reference | BP/IH | Market | Export |

**Chromatographic conditions:**

| | |
|---|---|
| **Column** | Stainless steel column (25 cm × 4.6 mm) packed with octylsilyl silica gel for chromatography (5 µm) (Waters Symmetry C8 is suitable) or equivalent. |
| **Injection volume** | 10µl |
| **Column Temperature** | 30°C |
| **Detection** | 220 nm |
| **Flow rate** | 1.0 ml per min |

**Sample Sequence:**

| Solution | No. of injections |
|---|---|
| Blank | 1 |
| Solution (3) | 1 |
| Solution (2) | 6 |
| Solution (4) | 1 |
| Solution (5) | 1 |
| Placebo | 1 |
| Blank | 1 |
| Sample solution | 1 |
| Bracketing Solution (2) | 1 |

**Note:** Discard any peak with an area less than the area of principle peak in the chromatogram obtained with solution (4) (0.1%).

Identify Impurity A from the relative retentions with reference to Solifenacin Succinate (retention time about 22 minutes) are: impurity A- about 0.4, impurity 1- about 0.8; impurity I- about 1.1 and impurity B- about 2.5.

**System Suitability:** The test is not valid unless

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1. The test is not valid unless, in the chromatogram obtained with solution (3), the resolution between the peaks due to solifenacin succinate and impurity I is at least 2.0.
2. In the chromatogram obtained with solution (4), the signal-to-noise ratio is at least 20.
3. %RSD for area of the standard six replicate injections is NMT 5.0%.

**Calculation:**

$$\text{Impurity I (\%)} = \frac{At}{As\,(sol-2)} \times \frac{Spl.Wt}{25} \times \frac{1}{200} \times \frac{25}{Spl.Wt} \times 100$$

$$\text{Impurity 1 (\%)} = \frac{Ag}{As\,(sol-2)} \times \frac{Spl.Wt}{25} \times \frac{1}{200} \times \frac{25}{Spl.Wt} \times 100$$

$$\text{Unknown Impurity (\%)} = \frac{Ai}{As\,(sol-2)} \times \frac{Spl.Wt}{25} \times \frac{1}{200} \times \frac{25}{Spl.Wt} \times 100$$

Total Unknown Impurities (%) = Sum of all unknown impurities.

**Where,**

| | |
|---|---|
| At | = Peak area obtained due to Impurity I |
| Ag | = Peak area obtained due to Impurity 1 |
| Ai | = Peak area obtained due to unknown impurity in sample solution |
| As (Sol – 2) | = Peak area obtained in the solution-2 |
| Spl. Wt | = Weight of Sample |

Disregard any peak with an area less than the area of the principal peak in the chromatogram obtained with solution (4) (0.1%).

**12. Microbial Contamination:**

**Instrument** : NA

**Reagents** : NA

**Procedure:** Refer Microbiology SOP Number: SOP/QCM/039

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| Quality Control Department | Finished Product Standard Test Procedure | Effective Date | 25/01/2023 |
| | Product Name: Solifenacin Succinate Tablets BP 5 mg | | |
| | Reference: BP/IH | Market | Export |

**Revision History Record :**

| Revision No. | Details for Revision(s) with Change Control No. | Effective Date |
|---|---|---|
| 00 | New Standard Test Procedure | 10/02/2021 |
| 01 | Complies as per current British Pharmacopoeia. | 25/01/2023 |

**Remark:** As per new format SOP No. SOP/QCD/086

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